
AXOLOTL: GENERATION OF AREA FORECAST DISCUSSIONS FROM AI WEATHER MODELS

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ABSTRACT

AI weather models increasingly produce high-dimensional forecasts spanning many variables, atmospheric levels, locations and lead times. For downstream use, these numerical forecasts often need to be interpreted and transformed into concise, application-relevant information such as area forecast discussions (AFDs). Large language models (LLMs) provide a natural mechanism for this transformation, but require an effective means to connect to the rich forecast state produced by the weather model. We introduce Axolotl, a parameter-efficient framework that proposes to directly couple the internal latent representation of a pretrained AI weather model to an LLM to generate AFDs. Both model backbones remain frozen, with only lightweight projection and adaptation parameters learned. We compare this latent interface with alternatives based on text summaries of selected variables and 2D gridded fields output by weather models. We train and evaluate our system using more than one million paired weather states and forecast targets derived from National Weather Service Area Forecast Discussions. The latent coupling achieves the strongest aggregate performance on a multiple-choice meteorological interpretation task, reaching 89.3% accuracy on a held-out year and 83.2% on unseen regions. It also performs strongly on open-ended forecast generation where nearly 60% of the generated forecasts receive a score of 6 or 7 from an LLM-as-a-judge, while a learned weather-state-text evaluator classifies 94.5% of the generations as consistent with the provided weather state.

1 Introduction

AI weather models are rapidly changing how the future state of the atmosphere and wider Earth system can be predicted [Pathak et al., 2022, Bi et al., 2023, Lam et al., 2023, Chen et al., 2023a,b, Kochkov et al., 2024, Lang et al., 2024, Bodnar et al., 2025, Price et al., 2025, Allen et al., 2025, Babenko and Coauthors, 2026]. The output of most of these forecasting systems is a high-dimensional numerical forecast: a collection of gridded fields describing the predicted evolution of the atmosphere. For many applications, this numerical state is an intermediate representation in a longer information-processing pipeline rather than the final product. Meteorologists produce *area forecast discussions (AFDs)* that explain atmospheric conditions and their evolution; public agencies issue warnings and impact assessments; and users in sectors such as agriculture, energy, transport and health require information tailored to a

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particular location, decision or audience. The ultimate output of many practical pipelines is therefore textual, ranging from technical meteorological analysis to warnings, decision support and personalized forecasts.

AI weather forecasting presents an opportunity: the forecasting model can be connected directly to other AI systems for these downstream tasks. Large language models (LLMs) provide a particularly natural conduit for applications whose output is text. Their ability to reason with natural language and draw on broad pretrained knowledge creates the possibility of systems that translate numerical Earth-system predictions into regional forecasts, answer questions interactively about predicted conditions, explain the physical trends underlying a forecast, describe potential impacts, or generate information tailored to a particular user or decision. Moreover, learning the coupling could also reduce the engineering and computational cost of constructing many separate downstream products, avoid information bottlenecks introduced by hand-designed intermediate representations, and allow the same underlying forecast to support many different queries and users. This opportunity is particularly timely because weather forecasts are becoming increasingly information-rich, with higher spatial, temporal and vertical resolution, more variables, and larger ensembles. LLM integration offers one path forward for transforming this information for end users.

The first technical challenge studied in this paper is how to connect the AI forecasting model to the LLM. We investigate a direct interface through the weather model’s learned latent representation. This provides a high-capacity interface through which the LLM can access information represented by the forecasting model without requiring the relevant fields or diagnostics to be selected in advance, avoiding a hand-designed information bottleneck that may discard signals needed for downstream tasks that were not anticipated when the interface was constructed. We compare this approach with two alternatives: textual summaries of selected forecast statistics, which are compact and interpretable but require hand-selecting variables and statistics; and decoded gridded fields, which retain more spatial information but similarly require choosing which variables to expose and grow in size with the amount of forecast information. The latent interface therefore provides a more general learned connection between the forecasting model and downstream reasoning, and suggests a modular architecture in which specialized AI models communicate through learned representations rather than manually engineered intermediate products.

The second technical challenge addressed in this paper is evaluation. While the ability of an interface to communicate meteorological information can be tested in controlled tasks, evaluating open-ended forecast generation is considerably harder. Open-ended forecast text does not generally have a unique correct realization: two forecasts may communicate the same meteorological information in very different ways. Simple text-overlap metrics therefore provide only limited evidence that a generated forecast faithfully represents the supplied weather state. Evaluation by expert meteorologists is crucial, but expensive, hard to perform at scale, and difficult to make fully reproducible. Useful evaluation of these systems therefore requires methods that can separately test whether the interface communicates meteorologically relevant information and whether freely generated text remains consistent with the underlying forecast state.

To address these two challenges, we introduce Axolotl, a parameter-efficient framework that couples the latent representation of a pretrained AI weather model to a language model for regional meteorological interpretation and forecast generation. In Section 2, we describe the Axolotl system, including the data used and the network architecture. In Section 3 we survey previous work on automatically turning numerical weather predictions into text, organized by the weather state interface to the text generator. Section 4 details experimental results where we demonstrate that the direct latent interface outperforms baselines on two tasks: a multiple-choice task evaluates whether Axolotl can recover meteorologically relevant information from the weather state; and a generation task where Axolotl is prompted to produce free-form, open-ended regional forecasts containing numerical values and near-term trends. In Section 5, we discuss how direct coupling of AI weather models to LLMs might extend to richer probabilistic forecasts and more general downstream applications. Our main contributions are the following:

- We introduce Axolotl, a parameter-efficient framework for directly coupling the internal representation of an AI weather model to an LLM, enabling regional weather reasoning and natural-language forecast generation without first reducing the weather-model state to a hand-selected textual representation.
- We construct and make available a large benchmark containing more than one million paired AI-weather-model states and AFD targets spanning ten forecast regions over 2007–2022, and develop a multiple-choice evaluation task alongside an open-ended forecast-generation task.
- We compare three distinct interfaces between the AI weather model and the LLM—text summaries, decoded gridded fields and learned latent representations—together with zero-shot and no-weather controls, under both temporal and geographic generalization.
- We show that latent conditioning provides the strongest aggregate performance: 89.3% accuracy on a multiple-choice meteorological interpretation task for a held-out year and 83.2% on unseen regions, with 59.0% of open-ended generations receiving LLM-judge scores of 6–7 on a seven point scale and 94.5% classified as consistent with the supplied weather state by our learned evaluator.

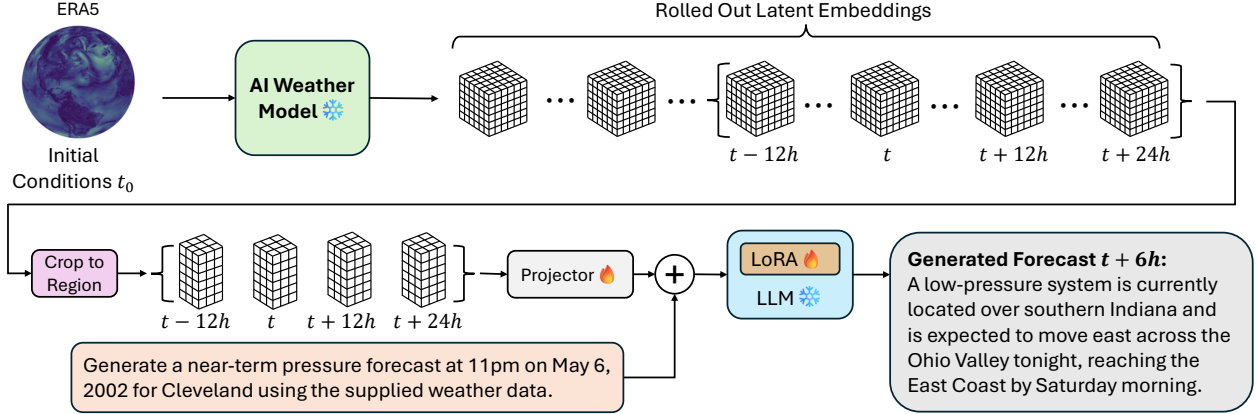


Figure 1: **Axolotl System.** From a set of ERA5 initial conditions, an AI weather model generates a set of global forecasts in the form of latent embeddings at four time points that have been rolled out between 1 and 7 days. The latents are then cropped to the region of interest and projected into the LLM prompt along with query text that contains the time, location, and variable for which to generate a forecast. The LLM then outputs the forecast as text. Note that t denotes the current time, the snowflake indicates that the network weights are frozen, while the flame indicates that the weights are learned.

2 Method

This section details the Axolotl system depicted in Figure 1, including the data used and the network architecture.

2.1 Data

Ground Truth Area Forecast Discussions (AFDs). Training Axolotl to generate area forecast discussions requires a dataset consisting of AI weather forecasts (inputs) and corresponding textual descriptions (targets). Ideally, one would have a large historical corpus of expert-written retrospective descriptions of the realized atmospheric state, which could be paired with forecasts made one or more days earlier. Such datasets are not generally available at scale. We therefore use the *near-term* sections of National Weather Service Area Forecast Discussions which detail the weather expected over the next 6 hours as a proxy. AFDs are particularly useful because they combine numerical information with qualitative descriptions and explanations of the meteorological processes underlying the forecast. Their near-term sections are written close to the target valid time and are strongly informed by current observations and the immediate evolution of the atmosphere. They therefore provide a practical proxy for the contemporaneous expert descriptions of the weather state that we would ideally use, while being available in large quantities over many years.

Importantly, the lead time of the weather-model input need not match the short forecast horizon of the textual target. A weather-model state may have been produced by rolling the model forward one or more days from an earlier initialization, while the corresponding AFD is written close to the valid time. We can therefore use these near-term descriptions as targets for learning to translate weather-model predictions made at lead times of several days. The AFD remains a forecast rather than a direct observation of realized weather, but its short lead time makes it a practical and accurate large-scale proxy for the retrospective descriptions we would ideally use.

For the ground truth targets, we download and collect meteorologist-authored AFDs from the Iowa Environmental Mesonet [Iowa State University, 2021] for a set of ten regions $\mathcal{R}$ (refer to Section A.1 for a list) in the United States and for the years 2007–2022. From each AFD, we extract only the text in the “NEAR TERM” section, which details the weather expected over the next 6 hours. In order to standardize the content of the AFDs, we then prompt a frontier LLM [Google DeepMind, 2025] to summarize the extracted near-term AFD text into information about four variables: $\mathcal{V} = \{\text{temperature, pressure, humidity, wind}\}$. Refer to Section A.2.1 for the prompt used and an example of the summarization. We use the variable summaries as the ground truth during training and evaluation. Pairing these targets with corresponding rolled-out predictions from the AI weather model produces more than one million training and test data points.

Dataset splits. We construct weather-state inputs paired with text-based variable forecast summaries over the period 2007–2022. The training split contains examples from 2007–2020, the validation split contains examples from 2021,

Table 1: Characteristics of the datasets used for the MCQ and Generation tasks. † indicates the regions used are OKX, HUN, APX, MHX, RLX, CLE, and BTV.

Split	Years	Regions	Samples	
			MCQ	Generation
Train	2007–2020	†	715,658	720,555
Validation	2021	†	4,158	14,280
Test (time split)	2022	†	71,077	71,645
Test (region split)	2007–2022	JAX, CAR, IND	237,401	240,091

and the time-generalization test split contains examples from 2022. These three splits contain data from seven forecast regions: OKX, HUN, APX, MHX, RLX, CLE, and BTV. To evaluate generalization across regions, we additionally construct a region-generalization test set containing examples from JAX, CAR, and IND over the complete 2007–2022 period. No examples from these three regions are used during training or validation. The characteristics of the datasets used for both the MCQ and generation tasks are shown in Table 1.

Weather State. In this work, we instantiate the AI weather model with Aurora [Bodnar et al., 2025], a pretrained foundation model for weather forecasting. We use Aurora to generate hourly weather forecasts for each of the ten regions $\mathcal{R}$ over the years 2007–2022, rolled forward by 1, 3, 5, and 7 days. Unless stated otherwise, references to the weather model in our experiments will refer to Aurora. We note that Axolotl is not specific to Aurora as it can be used with other AI weather models with suitable forecast outputs or internal representations. The weather model is initialized with ERA5 [Hersbach et al., 2020] reanalysis data at 0.25 degrees. Our goal is to generate area forecast discussions, a text-based forecast that describes how the weather evolves over a period of time, rather than an instantaneous snapshot. As a result, we require forecasts for multiple time points. To align with the time of a ground-truth variable summary, and allow the system to discern temporal changes in a forecast, for each example time point t , we train and evaluate on generated forecasts at $\mathcal{T}_t = \{t - 12, t, t + 12, t + 24\}$ hours. We generate forecasts for $\mathcal{T}_t$ as follows: for each $d \in \{1, 3, 5, 7\}$, initialize the AI weather model at d days before $t - 12$ and autoregressively roll it out in 6 hour steps for d days, and then an additional 36 hours of rollout to capture predictions at times t , $t + 12$, and $t + 24$ hours. Most AI weather forecasting models generate outputs in the form of gridded data for each variable (e.g. 2 m temperature) and atmospheric level. Some also allow the extraction of the latent state of all the variables, before they are decoded into gridded data [Bodnar et al., 2025].

Axolotl supports three configurations of the weather state for coupling to the LLM: text-based variable forecast summaries of the mean of the gridded variables that discard variation across the forecast region (*Statistics*); a predefined set of 2D gridded weather variables output from the AI weather model (*Gridded*); as well as latent embeddings extracted from the AI weather model prior to decoding into gridded variables and as a result contain the complete weather state generated by the model (*Latents*). These weather states allow us to ask whether communicating through the weather model’s internal representation provides a superior performing alternative to interfaces based on explicitly selected variables. Refer to Section A.3 for further details for weather state configurations.

2.2 Network Architecture

Our approach trains a single multi-task model across multiple regions, enabling transfer to held-out regions with limited or no training data. For a given time t , the AI weather model produces forecasts at all times in $\mathcal{T}_t$. These forecasts can be represented either as gridded physical variables or as latent embeddings. The gridded outputs are cropped to a $4^\circ \times 4^\circ$ region of interest, providing local atmospheric context for the target region. An analogous crop can be applied to Aurora’s latent representation because its latent tensor retains an explicit spatial organization: Aurora’s encoder and decoder operate locally, so each position in the latent grid corresponds to a geographical neighborhood in the decoded latitude–longitude grid. We therefore select the latent grid positions corresponding to the same $4^\circ \times 4^\circ$ region while retaining all atmospheric-level and feature dimensions. More generally, spatially organized latent representations are common in grid-based AI weather models that must decode their internal states back onto geographically indexed output grids, although the precise mapping and cropping procedure are architecture-dependent. The cropped gridded data or latent embeddings are then mapped into the LLM prompt with a lightweight projector consisting of a single linear layer with prefix and postfix RMSNorm [Zhang and Sennrich, 2019] functions. The projected vectors replace the embeddings of designated placeholder tokens in the LLM prompt. For the *Statistics* configuration, each variable is represented by its spatial mean at each input time and integrated directly into the LLM prompt as text. The LLM [Google DeepMind, 2026a] is then prompted to generate a near-term forecast for a selected region $r \in \mathcal{R}$ and time t using weather data in the form of either *Statistics*, *Gridded*, or *Latents*. Sample prompts are provided in Ap-

pendix A.2. The only learnable parameters are the LoRA weights in the otherwise frozen LLM and, where applicable, the lightweight projector. Consequently, fewer than 0.5% of the model parameters are trainable in our experiments.

3 Related Work

Automatically turning numerical weather predictions into text has a long history, and progress in both LLMs and AI weather models has renewed interest in the problem. We organize prior work by the weather state interface to the text generator: through hand-engineered or serialized summaries, through rendered or gridded fields processed by a vision encoder, or through learned embeddings taken from a weather model.

3.1 Rule-based and statistical forecast text generation

Glahn [1979] described one of the earliest operational systems for the U.S. National Weather Service that automates the production of public weather forecasts in tabular format as well as in natural language with a variable degree of complexity. FoG [Goldberg et al., 1994] converts weather maps into forecast text using rules and natural language generation for Environment Canada. SCRIBE [Verret et al., 1997] produced bilingual public forecasts from matrices of weather elements produced at an ensemble of 600 stations across Canada. Reiter et al. [2005] built the SUMTIME-MOUSAM weather-forecast generator to use consistent data-to-word rules, which avoided words which were only used by a few people, and words which were interpreted differently by different people. Belz [2008] used probabilistic generation-space methods to create weather forecast generators that produced outputs that were judged to have higher quality than forecasts written by experts. GALiWeather [Ramos-Soto et al., 2015] used fuzzy linguistic descriptions to produce short-term textual forecasts for 315 municipalities in Galicia from operational prediction data. In the natural language processing community, the WeatherGov corpus [Liang et al., 2009], which aligns NWS forecast text with the underlying forecast records, became a standard data-to-text benchmark and supported both probabilistic [Angeli et al., 2010] and neural encoder-aligner-decoder [Mei et al., 2016] generators. Common to these approaches is that the input is a small, preselected set of variables at points or over regions, and the content-selection logic must be designed or learned for that representation. They are the conceptual predecessors of the *Statistics* interface described in Section 2.

3.2 Language models operating on serialized forecast data

A natural modern replacement for hand-written realization rules is to serialize forecast data as text and prompt an LLM, in a similar fashion to our *Statistics* baseline. Gordon [2024] and Gordon and Cowling [2026] describe a system that queries a service providing weather forecast data as structured text [Open-Meteo.com, 2026], cleans and formats the retrieved data, and prompts a frontier LLM to write a human-friendly forecast. The Hierarchical AI-Meteorologist [Sukhorukov et al., 2025] serializes hourly point forecasts together with 6-hourly and daily aggregates and climatological normals, and asks an LLM agent to produce a narrative accompanied by summary keywords and a table-grounded evidence block intended to make the report verifiable; the system relies entirely on in-context prompting without task-specific training and is demonstrated through case studies.

Rather than serializing data directly, agentic systems allow the LLM to retrieve and compute the quantities it needs. Zephyrus [Varambally et al., 2026] provides a sandboxed Python environment exposing WeatherBench 2 ERA5 data [Rasp et al., 2024], a geolocation tool, the Stormer forecasting model [Nguyen et al., 2024], and an intermediate-complexity atmospheric simulator, and has an LLM write, execute and iteratively refine code to answer questions ranging from data lookup and forecasting to extreme-event detection, counterfactual reasoning and forecast-discussion writing. On its accompanying benchmark of over 2,000 questions spanning 46 task types, the agents substantially outperform text-only LLMs on retrieval and numerical questions, but perform similarly to them on the hardest tasks, and generating textual weather reports remains the weakest capability. Similar pipelines are now deployed at consumer scale: forecast summaries in Google’s Pixel Weather application are generated by a language model from the structured output of an AI forecasting system [Google, 2026].

3.3 Vision-language models over weather fields

A second family of approaches renders weather fields as images and processes them with vision-language models (VLMs). Lawson et al. [2025] examined the ability of GPT-4V to interpret meteorological imagery and to support forecast communication. WeatherQA [Ma et al., 2024] pairs more than 8,000 sets of Storm Prediction Center mesoscale-analysis parameters, surface observations and radar reflectivity images with expert-written mesoscale discussions, and evaluates VLMs on multiple-choice identification of the affected area, classification of the urgency and severity of current weather conditions, as well as generating mesoscale discussions, with the strongest model performance falling well short of human forecasters. ClimateQA [Chen et al., 2025] frames extreme weather event detection

as visual question answering over reanalysis-derived meteorological heatmaps. CLLMate [Li et al., 2024] pairs ERA5 fields with news reports of weather and climate events to pose event forecasting as a multiple-choice task.

Several recent works instead generate general-purpose forecast text from gridded data. Steele et al. [2025] fine-tune a VLM to write the UK Shipping Forecast directly from video-encoded gridded forecast fields, and compare it with an LLM given a rules-based textual representation of the same data; the VLM showed promise for visually driven elements such as visibility and sea state but was overall less consistent than the text-based approach. SynopticBench [Higgins et al., 2026] pairs forecast images of 500-hPa geopotential height, 2-m temperature anomaly and 850-hPa wind, with filtered AFD text from 117 National Weather Service forecast offices, yielding approximately 1.37 million raw image–text pairs and approximately 457,000 training pairs after restricting to lead times of at most 48 hours and to sentences describing synoptic-scale pressure and temperature features. LoRA fine-tuning improves open VLMs over their base versions, and the authors propose a location-aware metric for synoptic phenomena (SPACE), showing that conventional text-overlap metrics and LLM judges can rank models that generate synoptic discussions very differently.

The closest work to ours is WeatherSyn [Zheng et al., 2026], which fine-tunes Qwen3-VL-8B to generate multi-day structured and constrained area forecast discussions for 31 U.S. cities from heatmaps of 12 single-level ERA5 variables [Hersbach et al., 2020] at the initialization time. Its targets are drawn from the synopsis sections of AFDs over 2017–2022, segmented into daily forecasts and constrained, through aspect-controlled prompting, to eight meteorological aspects. Training proceeds in three stages: supervised fine-tuning, rejection-sampling fine-tuning to increase lexical diversity, and direct preference optimization. Corpus construction uses a larger LLM for segmentation and claim extraction, with manual correction of mismatched samples, yielding approximately 6,300 image–report pairs before augmentation. Evaluation is based on categorical claims (for example, warming, high pressure or a cold front) rather than numerical values. Because the model receives only the initial atmospheric state, the language model must implicitly perform the forecast itself; the authors report further gains when operational HRES forecast fields are added as inputs, which supports conditioning the language model on the output of a dedicated forecasting model.

3.4 Learned embeddings from weather models

Aquilon [Varambally et al., 2025] is the most direct precursor to our latent interface. It encodes a single global ERA5 snapshot (69 channels at 1.5° resolution) using a pretrained weather model, either a U-Net trained for 6-hour forecasting or the Stormer transformer [Nguyen et al., 2024], projects the resulting activations with a two-layer MLP into 1,024 or 2,048 tokens, and fine-tunes LLaMA-3.1-8B with LoRA while also updating the encoder. Trained on approximately 440,000 templated question–answer pairs covering seven retrieval, forecasting and extreme-event tasks, the embedding-based models gave only modest gains over a fine-tuned text-only LLM, and a training-free program-aided baseline [Gao et al., 2023] performed best on most tasks. The authors identified more effective weather representations and the inclusion of temporal context as key open directions.

3.5 Positioning of Axolotl

Axolotl differs from this prior work in four main respects. First, it conditions the LLM on the internal latent state of a frozen AI weather model [Bodnar et al., 2025], rather than on decoded fields, rendered images or a reanalysis snapshot, and the latents are taken from rollouts of one to seven days at four valid times, so the LLM receives both forecast information and its temporal evolution. Second, its targets are free-form, expert-derived text that includes numerical values and near-term trends, rather than templated answers or categorical claims. Third, the training data are extracted and preprocessed automatically with high accuracy, and as a result do not require expert verification or manual correction, and comprise more than one million paired instances. This is substantially larger than the WeatherSyn and Aquilon datasets and comparable in scale to SynopticBench, from which it differs in using AI weather model forecast states and near-term rather than synoptic-scale targets. Fourth, we demonstrate the effectiveness of the latent interface used by Axolotl, by comparing it to gridded and textual-statistics interfaces in like-for-like experiments, which allows the effect of the interface itself to be assessed directly. Prior work focuses on a single interface in isolation.

4 Experiments

We evaluate Axolotl on two tasks that progress from controlled measurement to open-ended forecasting. The first is a multiple-choice question (MCQ) task where the correct forecast for a specified date, time, and location must be selected from four alternatives. This provides a clear and controlled way to test whether the model can identify and reason about important weather information from each weather-state representation. We then move to the more difficult and less constrained task of generating concise AFDs that are consistent with the ground-truth ones. Evaluation of

the generation task is more challenging, because forecasts can be correct even if they are worded differently from the reference AFD. Across both tasks, we evaluate the three weather-state input configurations introduced in Section 2: text-based variable forecast summaries (*Statistics*), gridded weather variables (*Gridded*), and latent representations (*Latents*) that are generated by the AI forecasting model. Our experiments assess two forms of generalization: generalization to a future year for regions observed during training and generalization to held-out forecast regions.

We use Gemma-4-E4B-it [Google DeepMind, 2026a] as the base LLM for all trained configurations and Gemma zero-shot baselines. The same optimization protocol is followed for all fine-tuned configurations. The base LLM parameters remain frozen; task adaptation is restricted to rank-16 LoRA [Hu et al., 2022] parameters and, for *Latents* and *Gridded*, the corresponding projector. For *Latents* and *Gridded*, these trainable components account for 0.47% of the total model parameters. We also use Gemma-4-31B-it [Google DeepMind, 2026a] as the LLM-as-a-judge. We additionally evaluate Gemini-3.7-Flash [Google DeepMind, 2026b] as a 0-shot frontier-model baseline. No parameters are trained for this configuration. We train the MCQ models for three epochs and the generation models for ten epochs using AdamW [Loshchilov and Hutter, 2019] with a learning rate of 10^{-4} and an effective batch size of 32. Note that we train and evaluate different models for each of the tasks, input configurations, and forecast horizons. Full optimization settings and compute hardware are reported in Section A.4.

4.1 Multiple-Choice Questions (MCQs)

The MCQ task provides a controlled first test of how effectively each interface communicates the forecast state to the language model. Unlike the generation task, every example has only one correct answer and can therefore be evaluated directly with accuracy. For each AFD and each day d we create an MCQ about each variable $v \in \mathcal{V}$. Each MCQ has one correct answer and three incorrect distractor answers. We take the AFD summary for each variable v and set that to the correct answer for the MCQ. We create distractor answers for each MCQ by searching for the same variable and region, but from AFDs 7–14 days earlier or later, and then choosing three AFD summaries selected at random from the search results. The distractors are selected in this way so that they contain realistic weather for that time of year, meaning that they cannot be ruled out purely on climatological or meteorological grounds. It is possible that, by chance, the distractors are also compatible with the true weather state, but we have not controlled for this as it is difficult to do so. The four answer choices are randomly shuffled for each example during both training and evaluation to prevent the model from exploiting systematic answer-position biases. The LLM is then prompted to choose which of the four options best corresponds to the supplied weather state for a given time t and region $r \in \mathcal{R}$. For this task, we report accuracy, defined as the percentage of examples for which the model selects the correct forecast statement from the four candidate answers. A typical MCQ instance is given in Figure 2; example system and user prompts are depicted in Section A.2.2.

4.2 Near-Term Regional AFD Generation

Having established performance in the controlled MCQ setting, we evaluate whether the same interfaces support open-ended forecast generation. For each variable $v \in \mathcal{V}$ in each AFD and in each day d , the LLM is prompted to generate a forecast for the variable v . Example system and user prompts are depicted in Section A.2.3. To evaluate how well the generated variable forecasts reflect the ground truth variable summaries, we use two approaches. The first approach is to use a second LLM as a judge [Zheng et al., 2023]. This second LLM is prompted to use a prespecified seven-point scale to rate how similar and consistent the generated variable forecast is to the ground truth variable summary. The system and user prompts for the LLM-as-a-judge are given in Section A.2.4.

The second evaluation approach is to train a model to differentiate whether the forecast for a variable v is well matched to a given input weather state, or whether the forecast is mismatched with the given weather state. We do this as follows: we use the same dataset as that used in the MCQ case, but modify the MCQs to be true–false questions. Instead of having one correct answer and three incorrect distractor answers as in the MCQ case, for the weather state at time t for region r , 50% of the time we choose the correct variable forecast answer and assign to it the label *True*. The other 50% of the time we choose one of the three incorrect forecast answers at random and assign the label *False*. We then train a discriminator model on this true–false dataset, expecting that it will learn to differentiate when the variable forecast is matched to the weather state and when it is not. Once the discriminator is trained, we can pass generated variable forecasts along with their corresponding weather states into the discriminator and record the pass rate (i.e. the percentage of *True* responses) as the evaluation metric. As a sanity check, we can do the same with the ground-truth variable forecasts, which should yield a high pass rate if the discriminator is an accurate model. As a further verification that this model is producing meaningful predictions, we pass in a *mismatched* test set that has an incorrect weather state data that does not correspond with the ground truth or generated variable forecasts. When evaluated, the mismatched dataset should have a low pass rate.

MCQ: Wind Forecast

Gaylord, MI • 27 April 2022, 12:00 local time • Wind

Weather Data

Local time	Temp.	Wind	Pressure	Humidity
Wed 27 Apr, 00:00	28°F	13 mph NW	1025 hPa	69%
Wed 27 Apr, 12:00	34°F	12 mph N	1028 hPa	47%
Thu 28 Apr, 00:00	29°F	7 mph N	1029 hPa	69%
Thu 28 Apr, 12:00	43°F	5 mph E	1029 hPa	39%

Which forecast statement best matches the weather data?

A WSW winds today will diminish tonight, becoming light overnight.

B A NW flow is driving lake-effect snow showers. Winds will diminish tonight.

C Gusty southwest winds, at times over 40 mph, will slowly subside tonight. Gusts up to and over 30 mph will continue at times, but will fall below advisory levels quickly this evening.

D Winds are light or calm early this morning. Easterly low-level winds will strengthen today, then diminish tonight.

Model prediction: B Correct

Figure 2: **MCQ sample.** An MCQ sample for the *Statistics* configuration. The model receives weather-state information at multiple time points and selects the forecast statement that best agrees with the supplied weather state. The correct answer and model prediction are both B.

4.3 Additional Configurations

In addition to the *Statistics*, *Gridded*, and *Latents* weather states, we include two control baselines: *0-shot* and *No-Weather*. The *0-shot* baseline allows us to measure the contribution of fine-tuning the LLM versus using the pretrained weights. The *No-Weather* control removes the weather state information from the prompt while retaining the target forecast region, latitude, longitude, and local timestamp. The purpose of this baseline is to measure the contribution of the weather data to the predictions that the LLM makes. This is important for two reasons: (i) Since the evaluation examples may predate the model’s pretraining cutoff, strong performance without a supplied weather state could reflect memorized historical forecasts, climatological knowledge, or linguistic cues rather than reasoning over the forecast at hand. (ii) [Gu et al., 2026] finds that frontier LLMs can guess the correct answer even with key inputs removed. For these additional configurations, we use two models for *0-shot* evaluation: the open-weight base LLM (Gemma-4-E4B-it) and the more capable Gemini-3.7-Flash frontier model that has internet access and tool-use disabled. Owing to the frontier model’s inference cost and the considerably larger region-split test set, we restrict Gemini-3.7-Flash evaluation to the time split. In the results, the suffix *-ft* denotes a fine-tuned model, whereas *-0* denotes a pretrained model evaluated without task-specific fine-tuning.

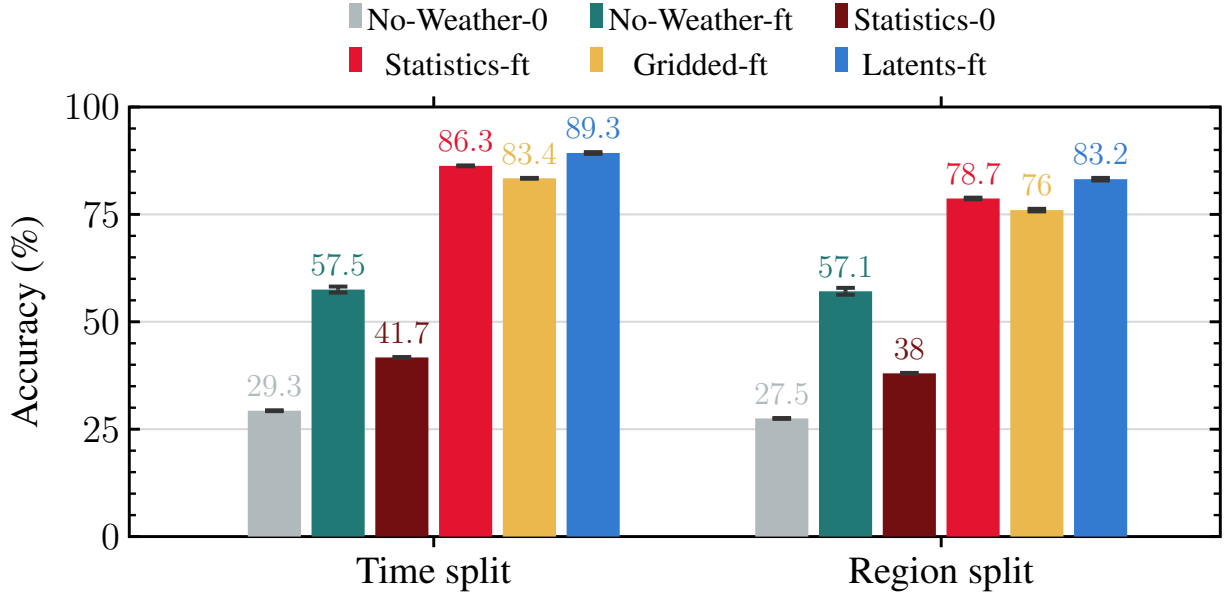


Figure 3: **MCQ Performance Across Input Configurations.** Accuracy (%) on the MCQ task using 1-day rollout data. Markers show mean accuracy and bars show standard errors over five random seeds for the fine-tuned base LLM configurations; all configurations utilize either a fine-tuned (“-ft” suffix) or a pretrained (“-0” suffix) base LLM.

4.4 MCQ Results

The results of the MCQ experiments are visualized in Figures 3 to 6 with tabulated versions in Tables A.1 and A.2. We draw five conclusions from these results, which we now discuss.

Weather-state conditioning substantially improves performance. Figure 3 shows that all fine-tuned weather-conditioned configurations substantially outperform *No-Weather-ft*. Specifically, *Latents-ft* improves by 31.8 percentage points on the time split and 26.1 percentage points on held-out regions over *No-Weather-ft*, demonstrating that the provided forecast state provides substantial information beyond location, seasonality, and dataset regularities. Fine-tuning also improves performance significantly compared to *0-shot* configurations.

Latents are best overall. Figures 3 and 4 show that *Latents-ft* ranks first on both splits, exceeding *Statistics-ft* by 3 percentage points in time and 4.5 percentage points on held-out regions, as well as outperforming the frontier LLM. In addition, *Statistics-ft* nevertheless outperforms *Gridded-ft*, showing that compact, physically meaningful summaries remain a strong interface despite discarding spatial information.

Fine-tuning the small base LLM outperforms the frontier LLM. Figure 4 indicates that (i) the frontier LLM substantially outperforms the base small LLM on the *Statistics-0* and *No-Weather-0* configurations, as expected; and (ii) the base LLM that has been fine-tuned on the latent outputs of the AI weather model outperforms the frontier LLM that is given *Statistics* by 17.7 percentage points, indicating the value of learning from a large task-specific dataset compared to the knowledge encoded in the pretrained frontier LLM. In the *No-Weather-0* case, the base LLM accuracy is close to chance, implying that the small LLM has not internalized weather knowledge to a significant degree during pretraining. However, the frontier LLM scores almost 60% in the same scenario indicating that the frontier model can exploit prior information from the location, date, candidate answers, or pretraining, even without an explicit weather state. Crucially, in the *Statistics-0* case, the frontier LLM raises its accuracy by a further 12 percentage points, indicating that the supplied weather information provides substantial signal beyond pretraining.

Performance remains strong as the rollout horizon increases. For each input configuration and rollout horizon, we fine-tune a separate model using only data from that horizon. From Figure 5 we see that *Latents-ft* has the highest accuracy at every horizon from 1 to 7 days, but its time-split accuracy decreases modestly from 89.5% at one day to 70.6% at seven days. As the rollout time horizon increases, a drop in the ability to correctly answer MCQs is expected due to the fact that the AI weather model forecast becomes less accurate. The other weather-conditioned systems degrade similarly. *No-Weather* remains approximately flat, as its responses are independent of the forecast.

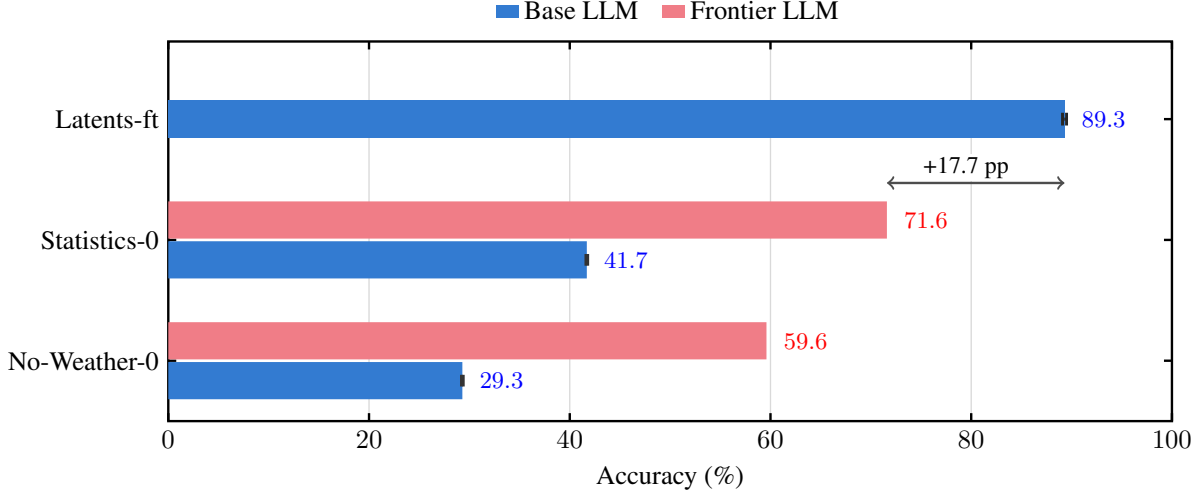


Figure 4: **Base and Frontier LLM Comparison.** MCQ accuracy comparison of zero-shot base and frontier LLMs with the fine-tuned base LLM system using latents (1-day rollouts). Error bars denote standard errors over five runs for the base LLM; frontier model results are from a single evaluation. All results are based on the time-split test set.

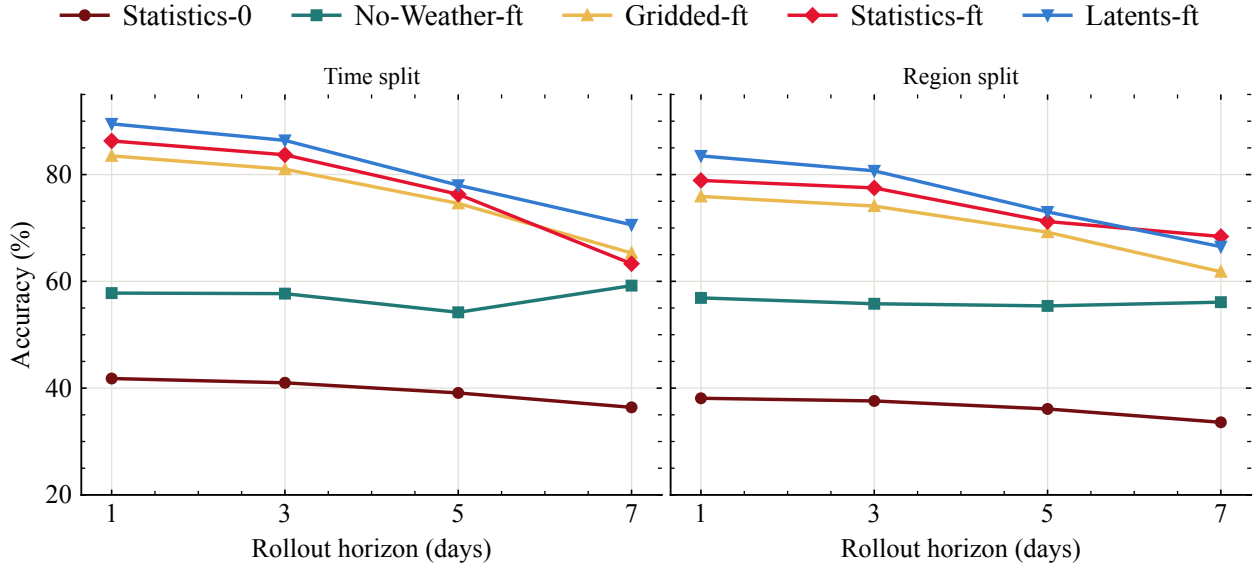


Figure 5: **MCQ Performance Across Rollout Horizons.** Accuracy (%) on the MCQ task as a function of forecast rollout horizon, for the time split (left) and region split (right). A separate model is fine-tuned from the same pretrained checkpoint of the base model for each input configuration and rollout horizon, using only data from the corresponding horizon.

Performance improves with longer training periods. For each training period in Figure 6, we fine-tune a separate *Latents-ft* model. Expanding the training period from 2020 alone to 2007–2020 improves accuracy by 12.8 percentage points in the time split and 14.4 percentage points in the region split. The particularly large improvement on held-out regions suggests that additional temporal diversity can also improve geographic transfer.

4.5 Generation results

Figure 8 shows the distributions of the LLM-as-a-judge scores for the generated variable forecasts for the time split. A similar figure for the region split can be found in Figure A.1 of the appendix. We see that the *Latents-ft* configuration produces the strongest overall score distribution, with 59.0% of the test set receiving a score of 6 or 7 on the seven-

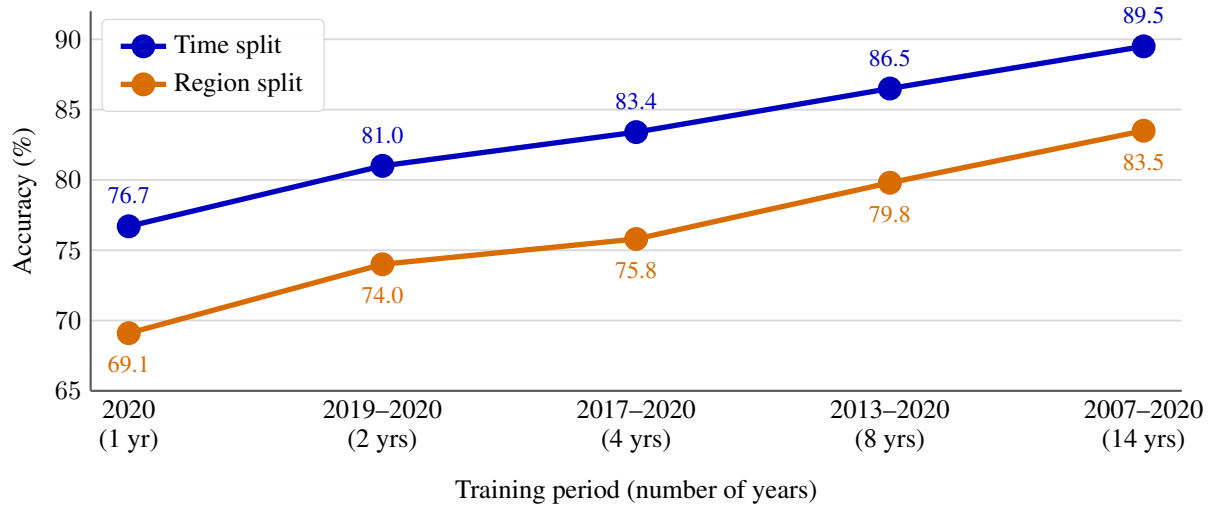


Figure 6: **MCQ Performance Across Training-Set Sizes.** MCQ accuracy as the amount of training data (1-day rollouts) increases for the time and region splits based on the *Latents-ft* configuration. A separate model is fine-tuned from the same pretrained checkpoint of the base model for each input training period, using only data from that period.

	Generated	Ground Truth
 Temperature	Overnight lows are expected to be in the mid 30s to mid 40s. Monday’s highs will be in the 50s to lower 60s, with Monday night lows in the 40s.	Lows will range from mid/upper 30s in eastern VT and the Adirondacks, and the lower/mid 40s elsewhere. Monday highs will be in the lower 60s in northern VT and mid/upper 50s further south. Monday night’s lows will be in the 40s.
 Pressure	A low-pressure system is currently located over southern Indiana and is expected to move east across the Ohio Valley tonight, reaching the East Coast by Saturday morning.	A vertically stacked low pressure system located over southern Indiana will continue to move eastward across the Ohio Valley this evening and to the mid-Atlantic region tonight.
 Humidity	The airmass is very dry, with precipitable water values around 0.25 inches. The airmass will remain very dry through tonight, with relative humidity values dropping to critical levels, potentially leading to elevated fire danger.	Exceedingly dry surface conditions are present. Source region dewpoints were generally in the 20s yesterday. Widespread critical level relative humidity values are anticipated later this morning and afternoon, contributing to elevated fire danger concerns due to low relative humidity.
 Wind	Damaging winds are the primary threat from any strong to severe storms that develop.	Damaging wind gusts are a continued main threat with strong to severe thunderstorms.

Figure 7: **Generation sample.** Sample variable forecast discussions that the LLM-as-a-judge rated as 7. The generated forecast discussions are from the *Latents-ft* configuration.

point scale. On the region split, we see analogous results, although all configurations register a lower percentage of 6 or 7 scores.

Sample variable generations and corresponding ground truth are given in Figure 7. The figure indicates that the *Latents-ft* configuration is able to generate responses that are highly consistent with the ground truth. Similar to the MCQ findings, the fine-tuned base LLM substantially outperforms the frontier LLM on the generation task. We see that the forecast-generation quality depends not only on the linguistic and reasoning capabilities of the LLM, but also on how effectively the weather state is communicated to it.

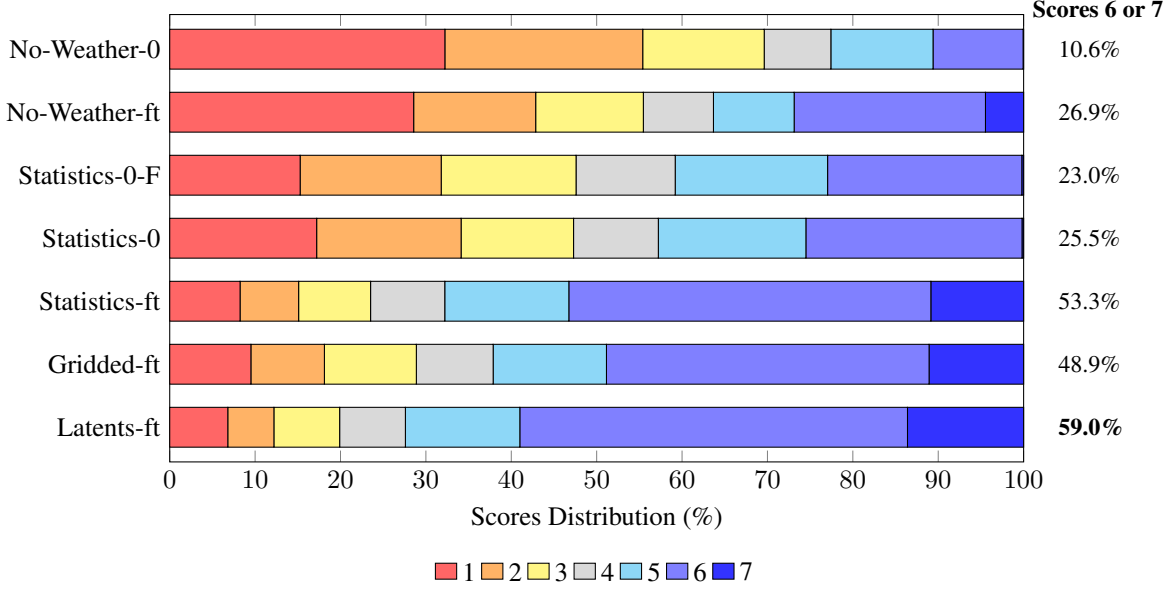


Figure 8: **Generation Score Distributions on the Time Split.** Distribution of LLM-as-a-judge scores across the seven generation configurations on the *time*-split test set using *1-day* rollouts. Values on the right report the percentage of generations receiving high scores (6 or 7). All configurations use either a fine-tuned or pretrained (-0 suffix) base LLM, except Statistics-0-F, which uses a frontier model zero-shot.

Figure 9 reinforces the evaluation results of the LLM-as-a-judge. The weather-state-text consistency discriminator evaluation indicates that the forecasts generated by the *Latents-ft* configuration have the highest pass rate at 94.5%, outperforming all other configurations. Interestingly, the pass rate for generations from *Latents-ft* exceeds the pass rate when the ground truth variable summaries are used in place of generations from Axolotl. This also highlights that its pass rate should be interpreted as weather-state-text consistency rather than as a direct measure of forecast quality; generated text may align more directly with features used by the discriminator than the AFD-derived reference summaries do. As expected, when the incorrect (i.e. mismatched) weather state is paired with a forecast generation, the pass rate falls dramatically for all configurations, providing credibility to the quality of the discriminator.

5 Discussion

In this final section, we summarize Axolotl performance, discuss its limitations, and make suggestions for future work.

5.1 Summary

We have presented Axolotl, a parameter-efficient framework for coupling the latent representation of an AI weather model directly to a language model. Our central finding is that an LLM can successfully exploit this latent representation for downstream meteorological reasoning and forecast generation. Across both temporal and geographic generalization, the latent configuration achieves the strongest performance, outperforming configurations based on selected decoded gridded variables and human-interpretable regional statistics. This is notable because the latent representation was learned for predicting the atmospheric state rather than explicitly constructed for the downstream language task. The result therefore suggests that the internal representations learned by AI weather models can provide useful general-purpose configurations to downstream models, and motivates exploring latent-state coupling for applications beyond forecast-text generation.

The strong performance of the *Statistics* configuration provides a second important finding. A relatively small set of carefully chosen meteorological summaries captures much of the information needed for the tasks studied here, but not all of it: the latent configuration consistently achieves stronger predictive performance. At the same time, the latent representation avoids having to specify in advance which variables, statistics, or spatial summaries will be relevant to a downstream task. Direct learned coupling therefore provides a convenient and extensible configuration through which the downstream model can automatically learn which components of the forecast state are useful, rather than requiring the configuration itself to be hand-crafted for each application. The frontier-model comparison reinforces

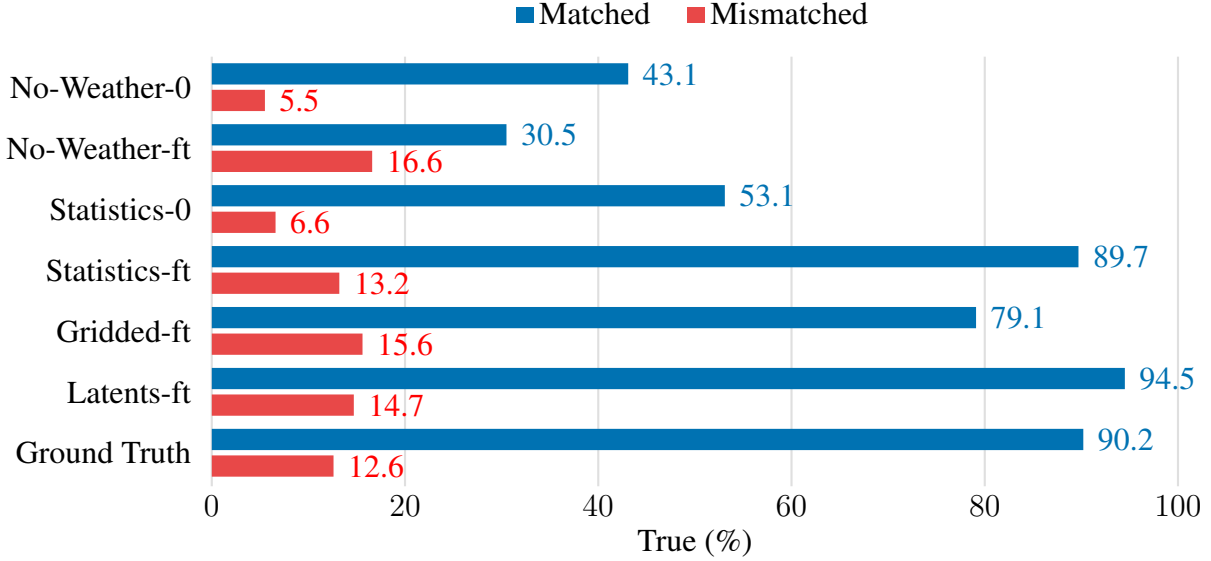


Figure 9: **Weather-State-Text Consistency Evaluation.** The blue Matched bars show the percentage of forecasts classified as consistent with their corresponding weather state. The red Mismatched bars show the same percentage after pairing forecasts with incorrect weather states; their lower values provide a sanity check. The results are based on the time-split test set and 1-day rollouts.

the importance of this configuration: despite using a substantially smaller language model, Axolotl with *Latents* outperforms the zero-shot frontier model supplied with textual weather statistics on both the MCQ and generation tasks.

5.2 Limitations

A main limitation is that the textual targets are derived from short-range AFD forecasts written close to the forecast valid time rather than retrospective expert descriptions of the realized atmosphere. Forecast errors in these targets therefore introduce label noise. If such errors are approximately unbiased, their main effect may be to increase the variance of the learned mapping rather than systematically bias it; nevertheless, datasets pairing historical forecasts with retrospective analyses, observations, or expert descriptions of realized weather would provide cleaner supervision and would make it easier to separate errors in the weather forecast from errors introduced by the language interface.

Our experiments also consider only two LLMs, Gemma-4-E4B-it and Gemini-3.7-Flash, and use a single AI weather model, Aurora, so further experiments are needed to establish how well the conclusions transfer across model families. The *Latents* and *Gridded* interfaces also differ in tokenization and information bottleneck: our *Gridded* interface compresses each selected variable into a single projected token, whereas the *Latents* representation is supplied as 256 tokens. The comparison therefore evaluates the complete interface designs rather than isolating the information content of the underlying representations. The experiments cover only ten forecast regions in the eastern United States and four weather variables. The generated forecasts are evaluated using automated metrics rather than expert assessment by meteorologists. Independent evaluation by meteorologists and experiments covering more regions, variables, and downstream tasks would provide stronger evidence that generated text faithfully represents the provided forecast state.

5.3 Future work

An important direction for future work is to evaluate Axolotl on more demanding tasks. The strong performance of the *Statistics* configuration suggests that the current tasks can often be addressed using compact regional summaries. Tasks that require reasoning over information distributed across many variables, spatial locations, or forecast trajectories would provide a stronger test of the benefits of latent representations.

Another promising extension to Axolotl is probabilistic forecasting. In many current probabilistic AI weather models, different stochastic noise realizations produce different forecast trajectories and consequently different latent states. A single latent realization therefore represents one possible future rather than the full predictive uncertainty of the forecast. Future architectures could instead construct latent representations that explicitly encode information across

an ensemble or predictive distribution, allowing uncertainty to remain available to the downstream language model rather than being collapsed into deterministic summaries before generation. An important question is then whether an LLM can faithfully interpret this uncertainty and communicate it in calibrated natural language.

Axolotl could also support multiple uses of the same underlying forecast state. Outputs could be tailored to meteorologists, farmers, emergency managers, or individual users, and the forecast could be used interactively rather than converted into a fixed product. The present work does not test such personalization, but the separation between the weather representation and textual query provides a natural architecture for doing so.

It is also possible to extend the learning setup itself. First, while Axolotl currently freezes the pretrained backbones and adapts the language model using parameter-efficient updates, future work could investigate updating a larger fraction or all of the downstream architecture. Second, training objectives could be extended so that downstream language or reasoning objectives influence the weather representation itself. Such joint optimization would need to preserve the forecasting accuracy and calibration of the underlying weather model while learning representations that are also useful to downstream tasks.

The forecast state could further be augmented with other sources of Earth-system information. Observations from satellites, radar, stations, and other sensors could be incorporated through additional learned configurations, allowing an LLM to combine coherent model predictions with local or rapidly updated observations and to reason explicitly about disagreement between them. Such components could sit within an agentic system that selects additional observations, forecasts, or specialist tools as required for a particular query. An important future test is whether configurations learned from data-rich regions can transfer to new regions, languages, or applications with relatively little local training data.

More broadly, we view Axolotl as an initial example of a compositional AI weather system in which specialized learned models communicate directly with one another. A weather foundation model provides a representation of the evolving Earth system; a language model provides reasoning and communication; and observations and other specialist models can contribute complementary information through learned configurations. The longer-term goal is not simply to attach a text generator to a numerical forecast, but to develop systems in which forecasting, uncertainty representation, reasoning, and communication operate as interacting components and can be optimized together.

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A Appendix

A.1 List of Area Forecast Regions

ID	Location	Split
OKX	Upton, NY	train (test-time)
HUN	Huntsville, AL	train (test-time)
APX	Gaylord, MI	train (test-time)
MHX	Newport, NC	train (test-time)
RLX	Charleston, WV	train (test-time)
CLE	Cleveland, OH	train (test-time)
BTV	Burlington, VT	train (test-time)
CAR	Caribou, ME	test-region
IND	Indianapolis, IN	test-region
JAX	Jacksonville, FL	test-region

A.2 Prompts

A.2.1 Area Forecast Discussion

Prompt A.1: Area Forecast Discussion Near Term Summarization Prompt

Act as a professional meteorologist and generate a detailed summary of the provided weather report into four categories: temperature, wind, pressure, and humidity. If no information is provided for any of the categories, the summary should say 'None'. Do not summarize anything related to adjustments to previous forecasts or if the forecast is on track. Output in strict JSON format with exactly four top-level keys (temperature, wind, pressure, and humidity) that contain only the summaries. Do not put keys for filename, summary, or model in the JSON file. There should be no descriptions of the task, no explanations, no additional information, no analysis, or thinking included in the JSON file.

Prompt A.2: Example Area Forecast Discussion Summarization

Near Term AFD for Huntsville, AL

- Issued at 4:02 AM CST Thu Jan 11, 2018

A deep upper low is moving through TX/OK with a surface low developing out to the east of it this morning. Southerly flow at the mid and upper levels are ushering in plenty of moisture as well as continued isentropic lift. This will keep scattered light rain showers in the forecast for most of the day. The "highest" rain chances look to be in the east with the slightly southeasterly flow at the surface. This warm south-southeasterly flow will push 850MB temps up to 10-11C, warmer than yesterday, and will push highs into the mid to upper 60s, some spots may even hit 70.

Variable Summaries Generated by Gemini 2.5 Flash

Temperature: High temperatures are forecast to reach the mid to upper 60s Fahrenheit, with some locations potentially reaching 70 degrees Fahrenheit. At the 850mb level, temperatures are expected to be 10-11 degrees Celsius, indicating warmer conditions compared to yesterday.

Wind: Southerly flow is observed at the mid and upper levels. At the surface, a slightly south-southeasterly flow, described as warm, is present.

Pressure: A deep upper low-pressure system is moving through Texas and Oklahoma, with a surface low-pressure system developing to its east this morning.

Humidity: Plenty of moisture is being ushered in by southerly flow, contributing to continued isentropic lift.

A.2.2 MCQ task

Prompt A.3: System Prompt for the MCQ Task

You are a meteorological forecast analyst. Your task is to match weather data with the most accurate forecast statement from multiple choices.

CORE PRINCIPLES:

1. Focus on the IMMEDIATE forecast period (next 0-6 hours from current time)
2. Analyze all meteorological parameters together: temperature trends, wind patterns, humidity levels, and pressure changes
3. Output ONLY the letter (A, B, C, or D) WITHOUT providing any explanation

Prompt A.4: User Prompt Instance of the *Latents* baseline for the MCQ Task

TASK: Select the single best forecast statement matching the weather data.

REGION: CLE - Cleveland, OH

LOCATION (lat, lon): (41.50, 278.31)

CURRENT LOCAL TIME: Friday, 31st of December, 2021 at 19:00

<start_of_weather_data>

<PH_ID> (repeated 256 times)

<end_of_weather_data>

- Variable focus: pressure

MULTIPLE-CHOICE OPTIONS:

- A) Building high pressure is anticipated to spread dry air through the region by Tuesday afternoon.
- B) Low pressure is currently located across Central Illinois and will continue to move northeast tonight. A subsequent wave of surface low pressure will track northeast through the Ohio Valley.
- C) General low pressure will affect the region for the next 12 to 24 hours. A trough will dig into the local area and deepen this afternoon, with a weak surface low taking shape across the southern fringe of the CWA.
- D) Surface high pressure will build across the Ohio Valley through the day Wednesday. The high pressure will center over the Ohio Valley Wednesday night.

OUTPUT ONLY the letter (A, B, C, or D):

Prompt A.5: User Prompt Instance of *Statistics* baseline for the MCQ Task

TASK: Select the single best forecast statement matching the weather data.

REGION: RLX - Charleston, WV

LOCATION (lat, lon): (38.35, 278.37)

CURRENT LOCAL TIME: Wednesday, 27th of April, 2022 at 13:00

Weather Data:

Format: - Local timestamp | Temperature | Wind | Pressure | Humidity

- Wednesday, 27th of April, 2022 at 01:00 | 42 F | 7 mph @ NorthWest | 1024 hPa | 69 %

- Thursday, 28th of April, 2022 at 01:00 | 40 F | 6 mph @ North | 1025 hPa | 69 %

- Thursday, 28th of April, 2022 at 13:00 | 55 F | 5 mph @ North | 1025 hPa | 36 %

CRITICAL ANALYSIS:

- Current condition: Wednesday, 27th of April, 2022 at 13:00 | 52 F | 13 mph @ NorthWest | 1024 hPa | 41 %

- Variable focus: temperature

MULTIPLE-CHOICE OPTIONS:

A) Radiational cooling is underway across the Central Appalachians, with temperatures falling into the 60s. Expect fairly strong radiational cooling overnight with clear skies, leading to values on the lowest end of guidance for typical cold spots and generally in the 25-50% quartile for remaining locations. Afternoon highs for Wednesday are forecast in the lower to mid-80s, with 60s in the mountains and 70s in mountain valleys.

B) Tonight, temperatures will fall below freezing across the northern half of the area, while the southern areas will see temperatures in the upper 30s. Highs on Thursday are expected to reach the 60s, with northern West Virginia mountains experiencing mid-40s. Frost Advisories and Freeze Warnings are in effect for parts of the Middle Ohio Valley and most of West Virginia tonight into Thursday morning.

C) Highs on Saturday are expected to be in the upper 50s to near 60 degrees. Overnight lows are forecast to be in the mid-40s.

D) Saturday's temperatures will be around 5 to 10 degrees cooler than today, with highs in the upper 50s to upper 60s in the lowlands, and 50s to low 60s in the mountains.

OUTPUT ONLY the letter (A, B, C, or D):

A.2.3 Generation task

Prompt A.6: System Prompt for the Generation Task

You are an expert meteorologist. Your task is to generate a near-term forecast for a weather variable (either temperature, pressure, humidity, or wind) for the specified date, time, and location, based on the supplied embedded weather data.

FORECAST RULES:

1. Length: The forecast must be descriptive, strictly between one and three sentences.
2. Numerical Data: You must include precise numerical data or numerical ranges with units derived from the weather data.
3. Trending Information: You should state any near-term trend of the variable over the next 6 hours.
4. Meteorological Cause: Where possible, provide a brief cause or explanation for the weather data and trend.
5. Output Format: Provide the forecast in the exact JSON structure requested.

Prompt A.7: User Prompt Instance of the *Latents* baseline for the **Generation** Task

```
<request>
  <task>Generate a near-term forecast for the specified weather variable at the specified date,
time, and location using the supplied weather data.</task>
  <variable>temperature</variable>
  <location>
    <region>BTV - Burlington, VT</region>
    <coordinates lat="44.48" lon="286.79">
  </location>
  <date_and_time>Thursday, 3rd of February, 2022 at 12:00</date_and_time>
  <weather_data>
    <semantic_context>The following block contains a continuous vector projection representing
atmospheric state variables for the specified region and time.</semantic_context>
    <latents>
      <PH_ID> (repeated 256 times)
    </latents>
  </weather_data>
</request>
```

Generate the forecast following the system rules and format the output as the following JSON object:

```
{
  "variable": "temperature",
  "forecast": "[Your forecast text here]"
}
```

Prompt A.8: User Prompt Instance of the *Statistics* baseline for the **Generation** Task

```

<request>
  <task>Generate a near-term forecast for the specified weather variable at the specified
  date, time, and location using the supplied weather data.
</task>
  <variable>temperature</variable>
  <location>
    <region>MHX - Newport/Morehead City, NC</region>
    <coordinates lat="34.72" lon="283.27" />
  </location>
  <date_and_time>Wednesday, 5th of October, 2022 at 06:00</date_and_time>
  <weather_data>
    <era5_stats>
      <format>Local timestamp | Temperature | Wind | Pressure | Humidity</format>
      <series>
        - Tuesday, 4th of October, 2022 at 18:00 | 62 F | 12 mph @ NorthWest | 1015 hPa | 53 %
        - Wednesday, 5th of October, 2022 at 18:00 | 66 F | 9 mph @ NorthWest | 1015 hPa | 59 %
        - Thursday, 6th of October, 2022 at 06:00 | 62 F | 5 mph @ NorthWest | 1017 hPa | 77 %
      </series>
      <current_condition>Wednesday, 5th of October, 2022 at 06:00 | 59 F | 11 mph @ NorthWest
                          | 1014 hPa | 62 %
    </current_condition>
  </era5_stats>
</weather_data>
</request>

Generate the forecast following the system rules and format the output as the following JSON
object:
{
  "variable": "temperature",
  "forecast": "[Your forecast text here]"
}

```

A.2.4 LLM-as-a-Judge

Prompt A.9: System Prompt for the Evaluation of the Generation Task

You are an expert meteorologist. Given two near-term weather forecast statements, your task is to score how well they capture the same essential meteorological information. In particular, you need to rate how similar and consistent the two statements are regarding the provided Focus Variable.

Key Principles:

- Focus on meteorological equivalence: Compare the content of the responses based upon the weather event details and characteristics rather than the exact times or locations described.
- Numerical comparability: Consider whether numerical values are similar and describe the same practical condition.
- Practical interpretation: Ask yourself: Would a meteorologist interpret these two statements as describing the same conditions? If yes, score high.
- Do not over-penalize minor differences: Variations in phrasing, relative versus absolute perspectives, or minor precision gaps should not lower scores if the essential meaning is consistent.
- Do not over-penalize based on the differences in length of one response versus the other. It is fine that one response is longer than the other as long as the two statements are consistent with each other.

Scoring Guidelines (1-7 Scale):

- 7 (Almost Identical): Responses describe the exact same phenomena with the same characteristics and numerical values. Practically identical forecast implications.
- 6 (Very Strong Alignment): Statements describe similar conditions with only slight differences. Same general trend or pattern, and numerical ranges overlap significantly.
- 5 (Strong Alignment): High alignment and consistency with comparable intensity (e.g., wind direction within a quadrant, temperature within 5 degrees). Practical forecast implications are very similar.
- 4 (Moderate Alignment): Statements share some common elements but differ in important details or magnitudes (e.g., wind direction off by a quadrant, temperature off by up to 10 degrees).
- 3 (Weak-Moderate Alignment): Partially overlapping information with notable gaps. Similar enough to be related but different enough to matter practically.
- 2 (Poor Alignment): Statements describe different aspects or characteristics. Different contexts, minimal practical overlap, or numerical differences are considerable (e.g., wind direction/strength considerably off, temperature difference beyond 15 degrees).
- 1 (No Alignment): Statements are inconsistent with each other and describe completely different, unrelated, or contradictory weather phenomena (e.g., "gusty" vs "calm" winds, "surface high" vs "low pressure").

Prompt A.10: User Prompt Instance for the Evaluation of the **Generation** Task

Evaluate the consistency of the following two forecast statements based on this specific variable:
 <focus_variable>temperature</focus_variable>

<statement_1>
 Overnight lows are expected to be in the mid 20s to lower 30s. Tuesday's highs will be in the upper 30s to mid 40s, with lows Tuesday night in the mid 20s to lower 30s.
 </statement_1>

<statement_2>
 Well below normal temperatures are expected through Tuesday night. Wind chills will be near zero for the Monday night/Tuesday morning commute. Highs on Tuesday will be in the lower 20s to lower 30s, remaining below normal. Lows Tuesday night will generally be in the teens, with single digits in the northern Adirondacks and far northeast Vermont.
 </statement_2>

Follow the system rules and format the output as the following JSON object:

```
{
  "score": [Integer between 1 and 7],
  "reasoning": "[Provide a 1-2 sentence technical summary]"
}
```

A.2.5 Weather-State-Text Consistency DiscriminatorPrompt A.11: System Prompt for the Binary Evaluator of the **Generation** Task

You are a meteorological forecast analyst. Your task is to decide whether a forecast statement accurately matches the weather data.

CORE PRINCIPLES:

1. Focus on the IMMEDIATE forecast period (next 0-6 hours from current time)
2. Analyze all meteorological parameters together: temperature trends, wind patterns, humidity levels, and pressure changes
3. Output ONLY 1 (if the statement matches the data) or 0 (if it does not) WITHOUT providing any explanation

Prompt A.12: User Prompt for the Binary Evaluator of the **Generation** Task (*Latents-fi*)

TASK: Decide whether the forecast statement accurately matches the weather data.

REGION: RLX - Charleston, WV
 LOCATION (lat, lon): (38.35, 278.37)
 CURRENT LOCAL TIME: Saturday, 14th of May, 2022 at 13:00

<start_of_weather_data>
 <PH_ID> (repeated 256 times)
 <end_of_weather_data>
 - Variable focus: temperature

FORECAST STATEMENT:

Diurnal heating will contribute to shower and thunderstorm development through late Saturday evening and again on Sunday.

OUTPUT ONLY 1 (matches) or 0 (does not match):

A.3 Weather State Details

Gridded-ft. The *Gridded-ft* configuration represents the weather state using a set of 2D weather-variable fields. Each example is represented by a tensor $\mathbf{X} \in \mathbb{R}^{T \times V \times H_g \times W_g}$, where $T = |\mathcal{T}| = 4$ is the number of input times from $\mathcal{T}$, $V = 5$ is the number of weather variables (2m_temperature, 10m_u_component_of_wind, 10m_v_component_of_wind, mean_sea_level_pressure, specific_humidity), and $H_g = W_g = 16$ are the height and width of the regional grid. Thus, each example has shape $[T, V, H_g, W_g] = [4, 5, 16, 16]$. For each weather variable, we flatten the time and grid dimensions. Since $TH_gW_g = 4 \times 16 \times 16 = 1024$, the resulting representation has shape $[L, TH_gW_g] = [5, 1024]$. Each of the five vectors is projected into the language-model embedding space, producing one embedding per weather variable. These embeddings replace the embeddings at five designated $\langle PH_ID \rangle$ placeholder positions in the input sequence. The placeholder tokens therefore specify where the projected weather-state vectors are inserted; they do not themselves encode meteorological information.

Latents-ft. The *Latents-ft* configuration uses internal representations extracted from the Aurora [Bodnar et al., 2025] AI weather model before its predicted weather state is decoded into individual weather-variable fields. For each of the $T = |\mathcal{T}| = 4$ input times in $\mathcal{T}$, Aurora produces a latent tensor with dimensions corresponding to the number of atmospheric levels, height, width, and latent feature embedding size. The complete latent representation is $\mathbf{Z} \in \mathbb{R}^{T \times A \times H_\ell \times W_\ell \times E}$, where $A = 4$ is the number of atmospheric levels, $H_\ell = W_\ell = 4$ are the latent grid dimensions, and $E = 1024$ is the latent feature dimension. The latent spatial dimensions are smaller than the corresponding gridded-data dimensions, $H_g = W_g = 16$, because the Aurora latent representation has a spatial resolution of 1.0° , whereas the decoded gridded fields have a resolution of 0.25° . Consequently, cropping the same $4^\circ \times 4^\circ$ geographic region yields a 4×4 latent grid but a 16×16 decoded weather grid. Therefore, $\mathbf{Z} \in \mathbb{R}^{4 \times 4 \times 4 \times 4 \times 1024}$. We flatten the time, level, and grid dimensions while preserving the latent feature dimension; for a batch of size B , this produces a tensor of shape $[B, T A H_\ell W_\ell, E] = [B, 256, 1024]$. The resulting 256 latent vectors are projected into the language-model embedding space and inserted at 256 $\langle PH_ID \rangle$ placeholder positions in the input sequence.

In contrast to the *Statistics-ft* configuration, which provides a predefined set of human-interpretable variable summaries, and the *Gridded-ft* configuration, which provides selected decoded weather variables, the *Latents-ft* configuration provides internal AI forecasting model representations. These representations may encode information distributed across the weather state learned by the forecasting model, including information that is not explicitly retained in the selected decoded variables or their regional summaries.

A.4 Optimization details

We use Gemma-4-E4B-it [Google DeepMind, 2026a] as the language-model backbone for all trained configurations and Gemma 0-shot baselines. We additionally evaluate Gemini-3.7-Flash [Google DeepMind, 2026b] as a 0-shot frontier-model baseline; no parameters are trained for this configuration. For the fine-tuned Gemma-4-E4B-it configurations, the backbone parameters remain frozen. Task adaptation is performed through rank-16 LoRA parameters and, for configurations with continuous weather-state inputs, the corresponding projector. For the Latents-ft and Gridded-ft configurations, rank-16 LoRA adapters and the projector account for 37,502,976 trainable parameters out of 7,978,603,808 total parameters (0.47%).

For both tasks, we optimize the trainable parameters using AdamW with a learning rate of 10^{-4} and an effective batch size of 32. We apply a weight decay of 10^{-3} to the projector parameters and no weight decay to the remaining trainable parameters. Each MCQ configuration is trained for three epochs, whereas each generation configuration is trained for ten epochs.

All fine-tuning experiments are conducted on a Linux server with NVIDIA RTX PRO 6000 Blackwell GPUs, each with 96 GB of VRAM. Training and evaluating one MCQ model takes approximately 33 wall-clock hours, while training and evaluating one generation model, including LLM-as-a-judge evaluation, takes approximately 50 wall-clock hours. Both use four GPUs, a per-GPU batch size of 2, and four gradient-accumulation steps, giving an effective batch size of 32.

A.5 Additional MCQ Results

Tables A.1 and A.2 provide the exact numerical results corresponding to Figures 3 and 5, respectively. Table A.1 reports accuracy aggregated across rollout horizons, with uncertainty estimated over five random seeds. Table A.2 further decomposes performance by the number of days for which the AI weather model is rolled out before constructing the weather-state input.

Table A.1: Accuracy (%) for the MCQ task; all configurations utilize either a fine-tuned (“-ft” suffix) or a pretrained (“-0” suffix) base LLM except No-Weather-0-F and Statistics-0-F where a frontier model is used.

Baseline	Time Split	Region Split
No-Weather-0-F	59.6	NA
No-Weather-0	29.3 $\pm$ 0.1	27.5 $\pm$ 0.1
No-Weather-ft	57.5 $\pm$ 0.7	57.1 $\pm$ 0.8
Statistics-0-F	71.6	NA
Statistics-0	41.7 $\pm$ 0.1	38.0 $\pm$ 0.1
Statistics-ft	86.3 $\pm$ 0.1	78.7 $\pm$ 0.2
Gridded-ft	83.4 $\pm$ 0.1	76.0 $\pm$ 0.3
Latents-ft	89.3 $\pm$ 0.2	83.2 $\pm$ 0.3

Table A.2: Accuracy (%) for the MCQ task as a function of rollout horizon. Best result per column in **bold**.

Baseline	Time Split				Region Split			
	1	3	5	7	1	3	5	7
No-Weather-ft	57.8	57.7	54.2	59.2	56.9	55.8	55.4	56.1
Statistics-0	41.8	41.0	39.1	36.4	38.1	37.6	36.1	33.6
Statistics-ft	86.3	83.7	76.3	63.3	78.9	77.5	71.2	68.4
Gridded-ft	83.5	81.0	74.6	65.3	75.9	74.1	69.2	61.8
Latents-ft	89.5	86.4	78.0	70.6	83.5	80.7	73.0	66.5

A.6 Additional Generation Results

Figure A.1 shows the score distributions on the region split. Latents-ft attains the highest percentage of generations receiving scores of 6 or 7, following the same pattern as in Figure 8.

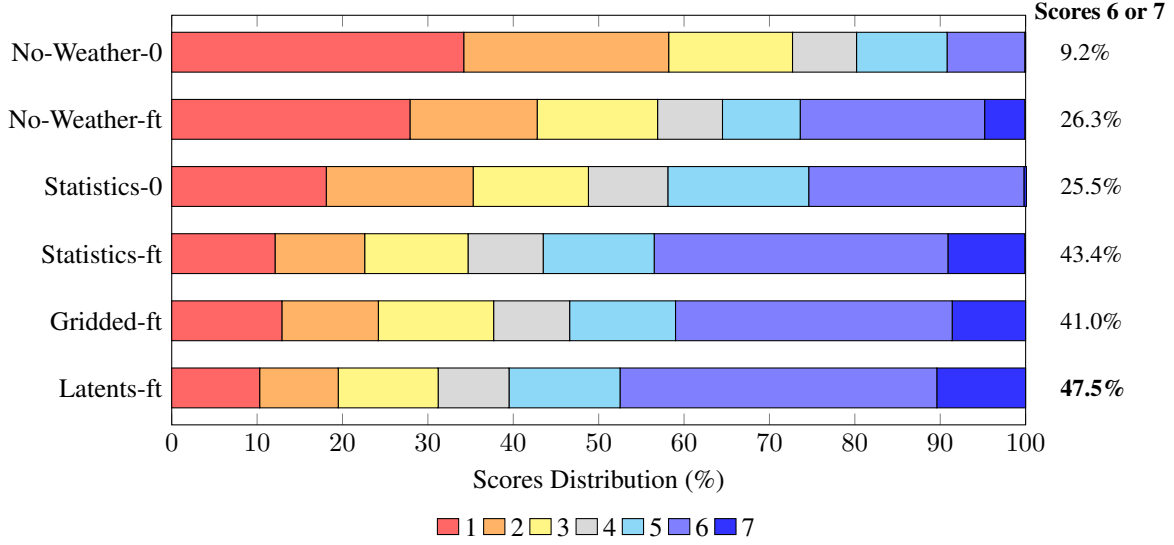


Figure A.1: **Generation Score Distributions on the Region Split.** Distribution of LLM-as-a-judge scores across the six generation configurations on the *region*-split test set using *1-day* rollouts. Values on the right report the percentage of generations receiving high scores (6 or 7). All configurations use either a fine-tuned or pretrained (-0 suffix) base LLM.